BFV Noise-Bound Extensions for Microsoft SEAL

Strict Bounds for RNS Key Switching, Relinearization, Rotation, Modulus Switching, and General Expressions

Introduction

The noise bounds for fresh encryption, negation, addition, subtraction, plain addition, plain subtraction, plain multiplication, and multiplication are established in Lemmas 9–17 of the Microsoft SEAL 2.3.1 manual (Appendix, strict infinity-norm bounds). This document extends this rigorous framework to operations implemented in SEAL's Residue Number System (RNS) arithmetic: key switching using an auxiliary special prime P, ciphertext relinearization, Galois rotation, and modulus switching. Finally, it establishes and proves an inductive noise bound for arbitrary BFV expressions of size N.

Notations

SEAL Notations

The notation adopted in this document adheres to that of the Microsoft SEAL 2.3.1 manual. The operations $\lfloor\cdot\rfloor$, $\lceil\cdot\rceil$, and $\lfloor\cdot\rceil$ denote rounding down, rounding up, and rounding to the nearest integer, respectively; when applied to a polynomial, they act coefficient-wise. The norm $\|\cdot\|$ denotes the polynomial infinity norm, defined for $a = \sum_{i=0}^{n-1} a_i X^i \in R \otimes \mathbb{Q}$ by $\|a\| = \max_{0 \le i < n} |a_i|$, while $\|\cdot\|_{\text{can}}$ denotes the canonical embedding norm. The reduction of an integer modulo t into the symmetric interval $[-t/2, t/2)$ is denoted by $[\cdot]_t$, and is applied coefficient-wise to polynomials. The base-2 logarithm is denoted by log, and the base-a logarithm by $\log_a$. Table 1 lists the parameters of the BFV scheme as defined in Table 1 of the SEAL manual. Furthermore, the ring expansion bound established in Claim 2 of the SEAL manual guarantees that for all $a, b \in R \otimes \mathbb{Q}$,

$$\|ab\| \le n \|a\| \|b\| .$$

Additional Notations

The derivations and proofs in this document employ the following additional notations:

- **RNS Primes and Modulus Sum**: The coefficient modulus is the product $q = \prod_{j=1}^{k} q_j$ of k pairwise coprime prime moduli q_j. Its prime sum is denoted by $q_\Sigma = \sum_{j=1}^{k} q_j$.
- **Special Auxiliary Prime** P: In RNS key switching, $P \ge 2$ denotes an auxiliary prime coprime to q, *i.e.*, $\gcd(P, q) = 1$.

Parameter	Description	Name in SEAL (if applicable)
q	Modulus in ciphertext space, of the form $q_1 \times \cdots \times q_k$, where q_i are prime	`coeff_modulus`
t	Modulus in plaintext space (plaintext modulus)	`plain_modulus`
n	A power of 2	`poly_modulus_degree`
$x^n + 1$	Polynomial modulus specifying the ring R	`poly_modulus`
R	The cyclotomic polynomial ring $\mathbb{Z}[x]/(x^n + 1)$	
R_a	The quotient ring $\mathbb{Z}_a[x]/(x^n + 1)$, with coefficients reduced modulo a	
w	Base into which ciphertext elements are decomposed during relinearization	
$\log w$	Decomposition bit count	`decomposition_bit_count`
ℓ	Evaluation key length parameter satisfying $\ell + 1 = \lfloor \log_w q \rfloor + 1$	
δ	Expansion factor in the ring R ($\delta \leq n$)	
Δ	Quotient on division of q by t, defined as $\lfloor q/t \rfloor$	
$r_t(q)$	Remainder on division of q by t, satisfying $q = \Delta t + r_t(q)$ with $0 \leq r_t(q) < t$	
χ	Error distribution over R	
B	Bound on the infinity norm of elements sampled from χ	

Table 1: Notation used throughout this document (reproduced from SEAL manual Table 1).

- **RNS Residues**: For any polynomial $c \in R_q$, its residue modulo q_j is denoted by $c^{(j)} = [c]_{q_j} \in [0, q_j)$.
- **Orthogonal CRT Reconstruction Factors**: For each $j \in \{1, \ldots, k\}$, the integer factor

$$\gamma_j = \frac{q}{q_j} \left[\left(\frac{q}{q_j} \right)^{-1} \right]_{q_j} \in \mathbb{Z}$$

  satisfies $\gamma_j \equiv 1 \pmod{q_j}$ and $\gamma_j \equiv 0 \pmod{q_i}$ for all $i \neq j$.
- **Rational Ring Extension**: The ambient rational algebra is denoted by $R \otimes \mathbb{Q} = \mathbb{Q}[x]/(x^n+1)$, which contains the scaled invariant noise polynomials.
- **Galois Automorphisms**: For an odd integer g coprime to $2n$, the map $\tau_g \colon R \to R$ is the ring automorphism defined by $X \mapsto X^g$.
- **Geometric Series Factor**: For any non-negative integer j, the factor $\frac{n^{j+1}-1}{n-1} = \frac{n^{j+1}-1}{n-1} = \sum_{i=0}^{j} n^i$ bounds the accumulation of powers of n.
- **Key Switching Operator**: The operator $\mathsf{KeySwitch}(c; \mathrm{evk})$ denotes the one-step RNS key switching procedure, mapping a ciphertext polynomial $c \in R_q$ to a pair of polynomials $(d_0, d_1) \in R_q^2$ using an evaluation key evk.

- **Ciphertext and Noise Designations**: Ciphertexts resulting from relinearization, rotation, modulus switching, plain addition, and plain subtraction are denoted by $\mathrm{ct}_{\mathrm{relin}}$, $\mathrm{ct}_{\mathrm{rot}}$, $\mathrm{ct}_{\mathrm{ms}}$, $\mathrm{ct}_{\mathrm{padd}}$, and $\mathrm{ct}_{\mathrm{psub}}$, with respective invariant noise polynomials v_{relin}, v_{rot}, v_{ms}, v_{padd}, and v_{psub}. The noise polynomials for addition, subtraction, multiplication, plain multiplication, and negation are denoted by v_{add}, v_{sub}, v_{mult}, v_{pmult}, and v_{neg}.
- **Recursive Expression Bound**: For an abstract FHE expression $\mathcal{E}$, the value $\mathcal{B}(\mathcal{E}) \in \mathbb{R}_{>0}$ denotes its computed invariant noise upper bound.

Conventions and Assumptions

The mathematical derivations in this document operate under the following formalized conventions and assumptions:

1. **Invariant Noise Convention**: Following Definition 1 of the SEAL manual, a ciphertext $\mathrm{ct} = (c_0, \ldots, c_j) \in R_q^{j+1}$ of size $j+1$ encrypts a plaintext message $m \in R_t$ with invariant noise $v \in R \otimes \mathbb{Q}$ under secret key $s \in R$ if

$$\frac{t}{q}\mathrm{ct}(s) = \frac{t}{q}\sum_{i=0}^{j} c_i s^i = m + v + at \in R \otimes \mathbb{Q},$$

   for some integer-coefficient polynomial $a \in R$. Decryption is successful whenever $\|v\| < 1/2$.

2. **Secret Key and Error Assumptions**: The secret key polynomial $s \in R$ has ternary coefficients, *i.e.*, $s \in R_3$, which guarantees $\|s\| = 1$. All fresh encryption and key-switching errors are sampled from distribution χ and satisfy $\|e\| \le B$.

3. **Algebraic and Notational Conventions**: Polynomial arithmetic is performed in the cyclotomic ring $R = \mathbb{Z}[x]/(x^n + 1)$. Ciphertext sizes are denoted exclusively by integers such as $j+1$, $M+1$, or K_{size}, reserving the symbol s exclusively for the secret key. The evaluation key length parameter ℓ from Table 1 is reserved for base-w decomposition; indices of RNS moduli are denoted exclusively by $j \in \{1, \ldots, k\}$.

Lemma 18: Key Switching (RNS form)

In SEAL, key switching from a secret key s' to the target secret key s is performed over the extended modulus qP. A key-switching key consists of k pairs $\mathrm{evk}[j] = (b_j, a_j) \in R_{qP}^2$ for $1 \le j \le k$, where

$$b_j = \left[-(a_j s + e_j) + P\gamma_j s'\right]_{qP}, \qquad a_j \xleftarrow{\$} R_{qP}, \quad e_j \leftarrow \chi, \quad \|e_j\| \le B.$$

For a ciphertext component $c \in R_q$, the procedure $\mathsf{KeySwitch}(c; \mathrm{evk})$ computes the linear combination over the integer lifts of the residues $c^{(j)} = [c]_{q_j} \in [0, q_j)$:

$$x_0 = \sum_{j=1}^{k} c^{(j)} b_j, \qquad x_1 = \sum_{j=1}^{k} c^{(j)} a_j,$$

and scales down by P using nearest-integer rounding:

$$\mathsf{KeySwitch}(c;\mathrm{evk}) = (d_0, d_1), \qquad d_0 = \left[\left\lfloor \frac{x_0}{P} \right\rceil\right]_q, \quad d_1 = \left[\left\lfloor \frac{x_1}{P} \right\rceil\right]_q.$$

By definition of nearest-integer rounding, $\lfloor x/P \rceil = (x - [x]_P)/P$, so that $\|\lfloor x/P \rceil - x/P\| \le 1/2$.

Lemma 18 (One-step key switching)**.** *Let $c \in R_q$ and $(d_0, d_1) = \mathsf{KeySwitch}(c;\mathrm{evk})$ for an evaluation key from s' to s. Then there exist an integer-coefficient polynomial $r \in R$ and an error polynomial $E \in R \otimes \mathbb{Q}$ such that*

$$d_0 + d_1 s = cs' + E + qr,$$

where

$$\|E\| \le \frac{nBq_\Sigma}{P} + \frac{n+1}{2} \le nBq_\Sigma.$$

Proof. By definition of rounding division by P followed by coefficient reduction modulo q, there exist rounding error polynomials $\delta_0, \delta_1 \in R \otimes \mathbb{Q}$ with $\|\delta_i\| \le 1/2$ and integer-coefficient polynomials $k_0, k_1 \in R$ such that

$$d_i = \left\lfloor \frac{x_i}{P} \right\rceil + qk_i = \frac{x_i}{P} + \delta_i + qk_i, \qquad i \in \{0, 1\}.$$

Lifting the modular definition of b_j from R_{qP} to R, there exist integer-coefficient polynomials $r_j \in R$ such that

$$b_j + a_j s = -e_j + P\gamma_j s' + qPr_j.$$

Evaluating $d_0 + d_1 s$:

$$\begin{aligned}
d_0 + d_1 s &= \frac{1}{P}(x_0 + x_1 s) + \delta_0 + \delta_1 s + q(k_0 + k_1 s) \\
&= \frac{1}{P}\sum_{j=1}^{k} c^{(j)}(b_j + a_j s) + \delta_0 + \delta_1 s + q(k_0 + k_1 s) \\
&= \frac{1}{P}\sum_{j=1}^{k} c^{(j)}\left(-e_j + P\gamma_j s' + qPr_j\right) + \delta_0 + \delta_1 s + q(k_0 + k_1 s) \\
&= -\frac{1}{P}\sum_{j=1}^{k} c^{(j)} e_j + \left(\sum_{j=1}^{k} c^{(j)}\gamma_j\right) s' + q\sum_{j=1}^{k} c^{(j)} r_j + \delta_0 + \delta_1 s + q(k_0 + k_1 s).
\end{aligned}$$

By the Chinese Remainder Theorem, since $\gamma_j \equiv 1 \pmod{q_j}$ and $\gamma_j \equiv 0 \pmod{q_i}$ for $i \ne j$, the modular congruence

$$\sum_{j=1}^{k} c^{(j)}\gamma_j \equiv c^{(j)} \equiv c \pmod{q_j}$$

holds for every $j \in \{1, \ldots, k\}$. Because the prime moduli q_j are pairwise coprime, their product divides the difference, establishing the existence of an integer-coefficient polynomial $K_{\mathrm{crt}} \in R$ such that

$$\sum_{j=1}^{k} c^{(j)}\gamma_j = c + qK_{\mathrm{crt}}.$$

Setting

$$E = -\frac{1}{P}\sum_{j=1}^{k} c^{(j)} e_j + \delta_0 + \delta_1 s, \qquad r = K_{\mathrm{crt}} s' + \sum_{j=1}^{k} c^{(j)} r_j + k_0 + k_1 s \in R,$$

gives the exact algebraic identity $d_0 + d_1 s = cs' + E + qr$.

To bound $\|E\|$, we apply the triangle inequality and the ring expansion bound from Claim 2 of the SEAL manual, namely $\|ab\| \le n \|a\| \|b\|$ for all $a, b \in R$:

$$\begin{aligned}\|E\| &\le \frac{1}{P} \sum_{j=1}^{k} \left\| c^{(j)} e_j \right\| + \|\delta_0\| + \|\delta_1 s\| \\ &\le \frac{1}{P} \sum_{j=1}^{k} n \left\| c^{(j)} \right\| \|e_j\| + \|\delta_0\| + n \|\delta_1\| \|s\| .\end{aligned}$$

Because $c^{(j)} \in R$ has integer coefficients in $[0, q_j)$, the inequality $\left\| c^{(j)} \right\| \le q_j$ holds. Substituting $\|e_j\| \le B$, $\|\delta_0\| \le 1/2$, $\|\delta_1\| \le 1/2$, and $\|s\| = 1$ yields:

$$\|E\| \le \frac{nB \sum_{j=1}^{k} q_j}{P} + \frac{1}{2} + \frac{n}{2} = \frac{nBq_\Sigma}{P} + \frac{n+1}{2}.$$

Finally, using $P \ge 2$ and $Bq_\Sigma \ge n + 1 \ge 2$, the inequality simplifies to:

$$\frac{nBq_\Sigma}{P} + \frac{n+1}{2} \le \frac{nBq_\Sigma}{2} + \frac{nBq_\Sigma}{2} = nBq_\Sigma.$$

□

Remark 1 (Algebraic Interpretation of Error E). The error polynomial E consists of two distinct components: the scaled sum of encryption errors $-\frac{1}{P} \sum_{j=1}^{k} c^{(j)} e_j$ and the rounding term $\delta_0 + \delta_1 s$. Dividing by P ensures that the key-switching noise is suppressed by the auxiliary modulus, eliminating the factor w present in base-w decomposition.

Lemma 19: Relinearization (RNS form)

Lemma 19 (Relinearization, RNS form)**.** *Let* $\mathrm{ct} = (c_0, \ldots, c_M)$ *be a ciphertext of size* $M + 1$ *encrypting* m *with noise* v*, and let* $\mathrm{ct}_{\mathrm{relin}}$ *of size* $N + 1$ *be obtained by relinearization modulo* q*, where* $2 \le N + 1 < M + 1$*. Then* $\mathrm{ct}_{\mathrm{relin}}$ *encrypts* m *and its noise* v_{relin} *satisfies*

$$\|v_{\mathrm{relin}}\| \le \|v\| + \frac{t}{q}(M - N) \left(\frac{nBq_\Sigma}{P} + \frac{n+1}{2} \right) \le \|v\| + \frac{t}{q}(M - N) nBq_\Sigma.$$

In particular, relinearizing down to size 2*, *i.e.*,* $N = 1$*, on a ciphertext of size* $K_{\mathrm{size}} = M + 1$ *yields*

$$\|v_{\mathrm{relin}}\| \le \|v\| + \frac{t}{q}(K_{\mathrm{size}} - 2) nBq_\Sigma.$$

Proof. Relinearization reduces the ciphertext size from $M + 1$ to $N + 1$ in $M - N$ successive elimination steps. We first analyze a single elimination step from size $K + 1$ down to size K, where $K \ge N + 1 \ge 2$. The input ciphertext is $(c_0, \ldots, c_K)$ with invariant noise v_K satisfying

$$\frac{t}{q} \sum_{i=0}^{K} c_i s^i = m + v_K + at,$$

for some integer polynomial $a \in R$. The highest-degree component c_K encrypts terms under key $s^K = s^{K-2} \cdot s^2$. Evaluating under secret key s, the term $c_K s^K$ is rewritten as $(c_K s^{K-2}) s^2$. Using the

relinearization key evk_K from s^2 to s, one computes $(d_0, d_1) = \mathsf{KeySwitch}(c_K; \mathrm{evk}_K)$. The updated ciphertext components are:

$$c'_i = c_i \quad (0 \le i \le K-3), \qquad c'_{K-2} = [c_{K-2} + d_0]_q, \qquad c'_{K-1} = [c_{K-1} + d_1]_q.$$

By Lemma 18, there exist an integer polynomial $r_K \in R$ and an error polynomial $E_K \in R \otimes \mathbb{Q}$ satisfying $\|E_K\| \le \frac{nBq_\Sigma}{P} + \frac{n+1}{2} \le nBq_\Sigma$ such that

$$d_0 + d_1 s = c_K s^2 + E_K + q r_K.$$

Lifting the reductions modulo q to R, there exist integer-coefficient polynomials $A_0, A_1 \in R$ such that

$$c'_{K-2} = c_{K-2} + d_0 + qA_0, \qquad c'_{K-1} = c_{K-1} + d_1 + qA_1.$$

Evaluating the decrypted value of the size-K ciphertext $\mathrm{ct}' = (c'_0, \ldots, c'_{K-1})$:

$$\begin{aligned}
\sum_{i=0}^{K-1} c'_i s^i &= \sum_{i=0}^{K-3} c_i s^i + (c_{K-2} + d_0 + qA_0)s^{K-2} + (c_{K-1} + d_1 + qA_1)s^{K-1} \\
&= \sum_{i=0}^{K-1} c_i s^i + (d_0 + d_1 s)s^{K-2} + q(A_0 + A_1 s)s^{K-2} \\
&= \sum_{i=0}^{K-1} c_i s^i + (c_K s^2 + E_K + q r_K)s^{K-2} + q(A_0 + A_1 s)s^{K-2} \\
&= \sum_{i=0}^{K} c_i s^i + E_K s^{K-2} + q\big(r_K + A_0 + A_1 s\big)s^{K-2}.
\end{aligned}$$

Scaling by t/q:

$$\begin{aligned}
\frac{t}{q}\mathrm{ct}'(s) &= \frac{t}{q}\sum_{i=0}^{K} c_i s^i + \frac{t}{q} E_K s^{K-2} + t\big(r_K + A_0 + A_1 s\big)s^{K-2} \\
&= m + v_K + at + \frac{t}{q} E_K s^{K-2} + t r'_K \\
&= m + \left(v_K + \frac{t}{q} E_K s^{K-2}\right) + (a + r'_K)t,
\end{aligned}$$

where $r'_K = \big(r_K + A_0 + A_1 s\big)s^{K-2} \in R$. Since $a + r'_K \in R$ has integer coefficients, ct' encrypts the message m, and its invariant noise is

$$v_{K-1} = v_K + \frac{t}{q} E_K s^{K-2}.$$

Because $K \ge 2$, the key-switching operation converts $c_K s^2$ directly to $d_0 + d_1 s$, eliminating the factor s^2 and replacing it with linear terms; for $K = 2$, $s^{K-2} = s^0 = 1$. Iterating across all $M - N$ elimination steps from $K = M$ down to $K = N + 1$:

$$v_{\mathrm{relin}} = v + \frac{t}{q} \sum_{K=N+1}^{M} E_K.$$

Applying the triangle inequality:

$$\|v_{\mathrm{relin}}\| \le \|v\| + \frac{t}{q} \sum_{K=N+1}^{M} \|E_K\| \le \|v\| + \frac{t}{q}(M-N)\left(\frac{nBq_\Sigma}{P} + \frac{n+1}{2}\right) \le \|v\| + \frac{t}{q}(M-N)nBq_\Sigma.$$

In the case of relinearization down to size 2, corresponding to $N = 1$, the number of elimination steps is $M - 1 = K_{\text{size}} - 2$, completing the proof. □

Remark 2 (Comparison with SEAL Lemma 12 and Implementation). Lemma 12 of the SEAL manual analyzes relinearization under base-w decomposition, resulting in an additive term proportional to $(\ell + 1)w$. In the RNS implementation of SEAL, the base-w decomposition is replaced by RNS decomposition across the prime factors q_j together with scaling by the special prime P. The factor $(\ell+1)w$ is thus replaced by $q_\Sigma = \sum_{j=1}^{k} q_j$. The reference implementation in `bfv_noise_bound.cpp` computes $(K_{\text{size}}-2)\frac{t}{q}nBq_\Sigma$, which matches Lemma 19 as a valid, slightly conservative upper bound.

Lemma 20: Galois Rotation

Lemma 20 (Galois rotation). *Let* $\mathrm{ct} = (c_0, c_1)$ *be a ciphertext of size* 2 *encrypting* m *with noise* v*, let* g *be an odd integer coprime to* $2n$*, and let* $\tau_g \colon R \to R$ *be the Galois automorphism defined by* $X \mapsto X^g$*. Let* evk_g *be the Galois key from* $s' = \tau_g(s)$ *to* s*. Let* $\mathrm{ct}_{\text{rot}} = ([\tau_g(c_0) + d_0]_q, d_1)$*, where* $(d_0, d_1) = \mathsf{KeySwitch}(\tau_g(c_1); \mathrm{evk}_g)$*. Then* ct_{rot} *encrypts* $\tau_g(m)$*, and its noise* v_{rot} *satisfies*

$$v_{\text{rot}} = \tau_g(v) + \frac{t}{q}E, \qquad \|v_{\text{rot}}\| \le \|v\| + \frac{t}{q}\left(\frac{nBq_\Sigma}{P} + \frac{n+1}{2}\right) \le \|v\| + \frac{t}{q}nBq_\Sigma.$$

Proof. The map τ_g is a ring automorphism of R and extends naturally to $R \otimes \mathbb{Q}$. Because $\gcd(g, 2n) = 1$, the exponent mapping $j \mapsto (jg \bmod 2n)$ defines a permutation of $\{0, \dots, 2n-1\}$. Using the cyclotomic relation $X^n \equiv -1$, each monomial X^{jg} maps to $\pm X^k$ for a distinct index $k \in \{0, \dots, n-1\}$. Thus τ_g permutes the polynomial coefficients up to sign changes, establishing that τ_g is an exact isometry on the infinity norm, *i.e.*, $\|\tau_g(a)\| = \|a\|$ for all $a \in R \otimes \mathbb{Q}$. Furthermore, $\tau_g(at) = \tau_g(a)t$, and τ_g maps integer-coefficient polynomials to integer-coefficient polynomials. Applying τ_g to the invariant noise relation $\frac{t}{q}(c_0 + c_1 s) = m + v + at$:

$$\frac{t}{q}(\tau_g(c_0) + \tau_g(c_1)\tau_g(s)) = \tau_g(m) + \tau_g(v) + \tau_g(a)t.$$

Applying Lemma 18 to the polynomial $c = \tau_g(c_1)$ with key evk_g from $s' = \tau_g(s)$ to s:

$$d_0 + d_1 s = \tau_g(c_1)\tau_g(s) + E + qr,$$

for some integer polynomial $r \in R$ and an error polynomial $E \in R \otimes \mathbb{Q}$ satisfying $\|E\| \le \frac{nBq_\Sigma}{P} + \frac{n+1}{2}$. Because Lemma 18 holds for an arbitrary source key s', the bound applies directly with $s' = \tau_g(s)$. Lifting the modular addition $[\tau_g(c_0)+d_0]_q$ from R_q to R, there exists an integer-coefficient polynomial $A_0 \in R$ such that $[\tau_g(c_0) + d_0]_q = \tau_g(c_0) + d_0 + qA_0$. Evaluating $\mathrm{ct}_{\text{rot}}(s)$:

$$\begin{aligned}
\frac{t}{q}\mathrm{ct}_{\text{rot}}(s) &= \frac{t}{q}(\tau_g(c_0) + d_0 + qA_0 + d_1 s) \\
&= \frac{t}{q}(\tau_g(c_0) + \tau_g(c_1)\tau_g(s)) + \frac{t}{q}E + t(r + A_0) \\
&= \tau_g(m) + \tau_g(v) + \tau_g(a)t + \frac{t}{q}E + t(r + A_0) \\
&= \tau_g(m) + \left(\tau_g(v) + \frac{t}{q}E\right) + (\tau_g(a) + r + A_0)t.
\end{aligned}$$

Since $\tau_g(a) + r + A_0 \in R$ has integer coefficients, $\mathrm{ct}_{\mathrm{rot}}$ encrypts $\tau_g(m)$, and its noise is represented by $v_{\mathrm{rot}} = \tau_g(v) + \frac{t}{q}E$. Taking the infinity norm and using the isometry $\|\tau_g(v)\| = \|v\|$:

$$\|v_{\mathrm{rot}}\| \le \|\tau_g(v)\| + \frac{t}{q}\|E\| \le \|v\| + \frac{t}{q}\left(\frac{nBq_\Sigma}{P} + \frac{n+1}{2}\right) \le \|v\| + \frac{t}{q}nBq_\Sigma.$$

□

Remark 3 (Scope and Multi-Step Rotations). In batched BFV encoding, the plaintext polynomial m encodes a matrix of plaintext slots, and the automorphism τ_g acts as a cyclic shift on the slots. Lemma 20 accounts for the exact cost of a single Galois key switch on a ciphertext of size 2. If a rotation by k steps is evaluated in software using h successive power-of-two automorphisms, the bound compounds to $\|v\| + h\frac{t}{q}nBq_\Sigma$. The single-key formula implemented in `bfv_noise_bound.cpp` corresponds to $h = 1$, which holds whenever the specific Galois key for the requested step is available.

Lemma 21: Modulus Switching

Lemma 21 (Modulus switching)**.** *Let* $\mathrm{ct} = (c_0, \ldots, c_j)$ *be a ciphertext of size* $j+1$ *modulo* $q = q_1 \cdots q_k$ *(*$k \ge 2$*) encrypting* m *with noise* v*. Write* $q = q'q_k$*, where* q_k *is the prime being dropped and* $q' = q/q_k = \prod_{i=1}^{k-1} q_i$ *is the target modulus. Let* $\mathrm{ct}_{\mathrm{ms}} = (c'_0, \ldots, c'_j)$ *with*

$$c'_i = \left\lfloor \frac{c_i}{q_k} \right\rceil \pmod{q'} = \frac{c_i - [c_i]_{q_k}}{q_k} \pmod{q'}, \qquad 0 \le i \le j.$$

Then $\mathrm{ct}_{\mathrm{ms}}$ *encrypts* m *modulo* q'*, and its noise* v_{ms} *satisfies*

$$\|v_{\mathrm{ms}}\| \le \|v\| + \frac{t}{2q'}\frac{n^{j+1}-1}{n-1} = \|v\| + \frac{t}{2q'} \cdot \frac{n^{j+1}-1}{n-1}.$$

Proof. For each $i \in \{0, \ldots, j\}$, let $r_i = [c_i]_{q_k} \in [-q_k/2, q_k/2)$ be the centered residue of c_i modulo q_k. Then $c_i - r_i$ is an exact integer multiple of q_k. Lifting to R, there exist integer-coefficient polynomials $A_i \in R$ such that

$$c'_i = \frac{c_i - r_i}{q_k} + q'A_i = \frac{c_i}{q_k} + \varepsilon_i + q'A_i,$$

where $\varepsilon_i = -r_i/q_k \in R \otimes \mathbb{Q}$. Because r_i is in the centered interval, $\|\varepsilon_i\| \le 1/2$. Evaluating $\frac{t}{q'}\mathrm{ct}_{\mathrm{ms}}(s)$ and using $q = q'q_k$:

$$\begin{aligned}
\frac{t}{q'}\mathrm{ct}_{\mathrm{ms}}(s) &= \frac{t}{q'}\sum_{i=0}^{j} c'_i s^i \\
&= \frac{t}{q'}\sum_{i=0}^{j}\left(\frac{c_i}{q_k} + \varepsilon_i + q'A_i\right)s^i \\
&= \frac{t}{q'q_k}\sum_{i=0}^{j} c_i s^i + \frac{t}{q'}\sum_{i=0}^{j}\varepsilon_i s^i + t\sum_{i=0}^{j} A_i s^i \\
&= \frac{t}{q}\mathrm{ct}(s) + \frac{t}{q'}\sum_{i=0}^{j}\varepsilon_i s^i + tA_{\mathrm{ms}},
\end{aligned}$$

where $A_{\text{ms}} = \sum_{i=0}^{j} A_i s^i \in R$. Substituting the definition of the invariant noise $\frac{t}{q}\text{ct}(s) = m + v + at$:

$$\frac{t}{q'}\text{ct}_{\text{ms}}(s) = m + \left(v + \frac{t}{q'} \sum_{i=0}^{j} \varepsilon_i s^i \right) + (a + A_{\text{ms}})t.$$

Because $a + A_{\text{ms}} \in R$ has integer coefficients, ct_{ms} encrypts m under the new modulus q', and the noise is represented by

$$v_{\text{ms}} = v + \frac{t}{q'} \sum_{i=0}^{j} \varepsilon_i s^i.$$

We now bound the infinity norm. By the triangle inequality:

$$\|v_{\text{ms}}\| \leq \|v\| + \frac{t}{q'} \sum_{i=0}^{j} \left\| \varepsilon_i s^i \right\|.$$

For $i = 0$, $s^0 = 1$, so $\left\|\varepsilon_0 s^0\right\| = \|\varepsilon_0\| \leq 1/2$. For $i \geq 1$, applying Claim 2 of the SEAL manual inductively with $\|s\| = 1$ yields $\left\|s^i\right\| \leq n^{i-1}$, and therefore:

$$\left\|\varepsilon_i s^i\right\| \leq n \left\|\varepsilon_i\right\| \left\|s^i\right\| \leq n \cdot \|\varepsilon_i\| \cdot n^{i-1} = n^i \|\varepsilon_i\| \leq \frac{1}{2} n^i.$$

Summing the finite geometric progression:

$$\sum_{i=0}^{j} \left\|\varepsilon_i s^i\right\| \leq \frac{1}{2} \sum_{i=0}^{j} n^i = \frac{1}{2} \frac{n^{j+1} - 1}{n - 1} = \frac{1}{2} \frac{n^{j+1} - 1}{n - 1}.$$

Therefore,

$$\|v_{\text{ms}}\| \leq \|v\| + \frac{t}{2q'} \frac{n^{j+1} - 1}{n - 1} = \|v\| + \frac{t}{2q'} \cdot \frac{n^{j+1} - 1}{n - 1}.$$

□

Remark 4 (Relation to Geometric Sum in Implementation). The geometric sum in Lemma 21 is identical to inequality (1) in Lemma 11 of the SEAL manual. In `bfv_noise_bound.cpp`, the helper `log_geometric_sum(n, j+1)` returns $\sum_{i=1}^{j+1} n^i = n\frac{n^{j+1}-1}{n-1}$. The implemented bound is therefore larger by a factor of n, representing a valid but conservative upper bound on the provable bound of Lemma 21.

Synthesis and General Expression Noise Bound

Synthesis of Noise Bounds

Table 2 synthesizes the strict infinity-norm noise bounds for all elementary BFV operations, combining the textbook results of the SEAL manual (Lemmas 9–17) and the RNS extensions derived in Lemmas 19–21.

Operation	Source	Invariant Noise Bound $\|v_{\text{out}}\|$	Size	Modulus
Elementary operations from SEAL 2.3.1 manual (Lemmas 9–17)				
Fresh encryption	Lemma 9	$\frac{r_t(q)}{q}\|m\| + \frac{tB}{q}(2n+1)$	2	q
Negation	Lemma 15	$\|v\|$	$j+1$	q
Addition	Lemma 10	$\|v_1\| + \|v_2\|$	$\max(j_1, j_2)+1$	q
Subtraction	Lemma 16	$\|v_1\| + \|v_2\|$	$\max(j_1, j_2)+1$	q
Plain addition	Lemma 14	$\|v\| + \frac{r_t(q)}{q}\|m_2\|$	$j+1$	q
Plain subtraction	Lemma 17	$\|v\| + \frac{r_t(q)}{q}\|m_2\|$	$j+1$	q
Plain multiplication	Lemma 13	$\|m_2\|_1 \|v\| \le N_{m_2}\|m_2\|\|v\|$	$j+1$	q
Multiplication	Lemma 11	$\left[(N_{m_1}+n)\|m_1\| + \frac{nt}{2}\frac{n^{j_1+1}-1}{n-1}\right]\|v_2\| + \left[(N_{m_2}+n)\|m_2\| + \frac{nt}{2}\frac{n^{j_2+1}-1}{n-1}\right]\|v_1\| + 3n\|v_1\|\|v_2\| + \frac{t}{2q}\frac{n^{j_1+j_2+1}-1}{n-1}$	j_1+j_2+1	q
Derived operations in RNS representation				
Relinearization	Lemma 19	$\|v\| + \frac{t}{q}(K_{\text{size}}-2)\,nBq_\Sigma$	2	q
Galois rotation	Lemma 20	$\|v\| + \frac{t}{q}\,nBq_\Sigma$	2	q
Modulus switching	Lemma 21	$\|v\| + \frac{t}{2q'}\frac{n^{j+1}-1}{n-1}$	$j+1$	$q' = q/q_k$

Table 2: Synthesis of strict BFV invariant noise bounds for elementary operations.

Formalization of an FHE Expression

An FHE computation is formally modeled as an Abstract Syntax Tree (AST) representing the evaluation of ciphertexts.

Definition (BFV Noise Expression of Size N)**.** A *BFV noise expression* $\mathcal{E}$ of size $N \in \mathbb{N}_{\ge 1}$, representing the number of operation nodes, is defined inductively:

1. **Base case ($N = 1$)**: A fresh encryption $\mathcal{E} = \mathsf{fresh}(m)$, where $m \in R_t$ is a plaintext message.
2. **Unary operations ($N = N_1 + 1$)**: If $\mathcal{E}_1$ is an expression of size N_1, then $\mathcal{E}$ is formed by one of:

$$\mathcal{E} \in \big\{\mathsf{negate}(\mathcal{E}_1),\ \mathsf{add_plain}(\mathcal{E}_1, m_2),\ \mathsf{sub_plain}(\mathcal{E}_1, m_2),\ \mathsf{multiply_plain}(\mathcal{E}_1, m_2),\ \mathsf{relinearize}(\mathcal{E}_1),\ \mathsf{rotate}(\mathcal{E}_1),\ \mathsf{mod_switch}(\mathcal{E}_1)\big\},$$

   where $m_2 \in R_t$ is a known plaintext polynomial.
3. **Binary operations ($N = N_1 + N_2 + 1$)**: If $\mathcal{E}_1$ and $\mathcal{E}_2$ are expressions of sizes N_1 and N_2 evaluated at the same modulus level, then:

$$\mathcal{E} \in \big\{\mathsf{add}(\mathcal{E}_1, \mathcal{E}_2),\ \mathsf{sub}(\mathcal{E}_1, \mathcal{E}_2),\ \mathsf{multiply}(\mathcal{E}_1, \mathcal{E}_2)\big\},$$

   is an expression of size N.

For each expression $\mathcal{E}$, its evaluation state is a triple $\mathcal{S}(\mathcal{E}) = (\mathcal{B}(\mathcal{E}),\ K_{\text{size}}(\mathcal{E}),\ q(\mathcal{E}))$, where $\mathcal{B}(\mathcal{E}) \in \mathbb{R}_{>0}$ is the computed noise upper bound, $K_{\text{size}}(\mathcal{E}) \in \mathbb{N}_{\ge 2}$ is the ciphertext size, and $q(\mathcal{E})$ is the current coefficient modulus.

Definition (Recursive Bound Function $\mathcal{B}(\mathcal{E})$)**.** The bound function $\mathcal{B}(\mathcal{E})$ is defined recursively matching the rules in Table 2:

$$\begin{aligned}
\mathcal{B}(\mathsf{fresh}(m)) &:= \frac{r_t(q)}{q}\,\|m\| + \frac{tB}{q}(2n+1),\\
\mathcal{B}(\mathsf{negate}(\mathcal{E}_1)) &:= \mathcal{B}(\mathcal{E}_1),\\
\mathcal{B}(\mathsf{add_plain}(\mathcal{E}_1, m_2)) &:= \mathcal{B}(\mathcal{E}_1) + \frac{r_t(q)}{q}\,\|m_2\|\,,\\
\mathcal{B}(\mathsf{sub_plain}(\mathcal{E}_1, m_2)) &:= \mathcal{B}(\mathcal{E}_1) + \frac{r_t(q)}{q}\,\|m_2\|\,,\\
\mathcal{B}(\mathsf{multiply_plain}(\mathcal{E}_1, m_2)) &:= \min(\|m_2\|_1\,,\ N_{m_2}\,\|m_2\|)\cdot\mathcal{B}(\mathcal{E}_1),\\
\mathcal{B}(\mathsf{add}(\mathcal{E}_1, \mathcal{E}_2)) &:= \mathcal{B}(\mathcal{E}_1) + \mathcal{B}(\mathcal{E}_2),\\
\mathcal{B}(\mathsf{sub}(\mathcal{E}_1, \mathcal{E}_2)) &:= \mathcal{B}(\mathcal{E}_1) + \mathcal{B}(\mathcal{E}_2),\\
\mathcal{B}(\mathsf{multiply}(\mathcal{E}_1, \mathcal{E}_2)) &:= \left[(N_{m_1}+n)\,\|m_1\| + \frac{nt}{2}\frac{n^{j_1+1}-1}{n-1}\right]\mathcal{B}(\mathcal{E}_2)\\
&\quad + \left[(N_{m_2}+n)\,\|m_2\| + \frac{nt}{2}\frac{n^{j_2+1}-1}{n-1}\right]\mathcal{B}(\mathcal{E}_1)\\
&\quad + 3n\mathcal{B}(\mathcal{E}_1)\mathcal{B}(\mathcal{E}_2) + \frac{t}{2q}\frac{n^{j_1+j_2+1}-1}{n-1},\\
\mathcal{B}(\mathsf{relinearize}(\mathcal{E}_1)) &:= \mathcal{B}(\mathcal{E}_1) + \frac{t}{q}(K_{\text{size}}(\mathcal{E}_1)-2)\,nBq_\Sigma,\\
\mathcal{B}(\mathsf{rotate}(\mathcal{E}_1)) &:= \mathcal{B}(\mathcal{E}_1) + \frac{t}{q}\,nBq_\Sigma,\\
\mathcal{B}(\mathsf{mod_switch}(\mathcal{E}_1)) &:= \mathcal{B}(\mathcal{E}_1) + \frac{t}{2q'}\frac{n^{K_{\text{size}}(\mathcal{E}_1)-1+1}-1}{n-1},
\end{aligned}$$

where $j_i = K_{\text{size}}(\mathcal{E}_i) - 1$, and $q' = q/q_{\text{last}}$. Whenever the underlying plaintext messages m_1, m_2 are unknown intermediate ciphertexts, universal upper bounds $\|m_i\| \le \lfloor t/2 \rfloor$ and $N_{m_i} \le n$ are substituted, yielding $(N_{m_i}+n)\,\|m_i\| \le 2n\lfloor t/2 \rfloor \le nt$. For plain addition and subtraction, the exact theoretical bound retains the additive remainder term $\frac{r_t(q)}{q}\,\|m_2\|$ from Lemma 14; its connection to the zero-growth simplification in `bfv_noise_bound.cpp` is clarified in Remark 6.

Lemma 22: Soundness of the Expression Noise Bound

Lemma 22 (Soundness of the Expression Noise Bound)**.** *Let $\mathcal{E}$ be any well-formed BFV noise expression of size $N \in \mathbb{N}_{\ge 1}$. Let $\mathrm{ct}_{\mathcal{E}}$ be the ciphertext resulting from evaluating $\mathcal{E}$, and let $v(\mathcal{E})$ be its invariant noise. Then:*

$$\|v(\mathcal{E})\| \le \mathcal{B}(\mathcal{E}). \tag{1}$$

Furthermore:

(i) Decryption Correctness*: If $\mathcal{B}(\mathcal{E}) < 1/2$, then $\mathrm{ct}_{\mathcal{E}}$ decrypts correctly to the intended message $m(\mathcal{E}) \in R_t$:*

$$\mathrm{Decrypt}(\mathrm{ct}_{\mathcal{E}}) = m(\mathcal{E}).$$

(ii) Noise Budget Guarantee*: The remaining invariant noise budget of* $\mathrm{ct}_{\mathcal{E}}$ *satisfies:*

$$\mathrm{Budget}(\mathrm{ct}_{\mathcal{E}}) \geq -\log_2(2\mathcal{B}(\mathcal{E})) \quad \textit{bits.}$$

Proof. We prove (1) by mathematical induction on the size N of the expression $\mathcal{E}$.

Base Case ($N = 1$): The only well-formed expression of size 1 is a fresh encryption $\mathcal{E} = \mathsf{fresh}(m)$. By Lemma 9 of the SEAL manual, the invariant noise v of a fresh encryption satisfies:

$$\|v(\mathcal{E})\| \leq \frac{r_t(q)}{q}\|m\| + \frac{tB}{q}(2n+1) = \mathcal{B}(\mathcal{E}).$$

The base case holds.

Inductive Hypothesis: Assume that for all well-formed expressions $\mathcal{E}'$ of size k with $1 \leq k < N$, the evaluated ciphertext satisfies $\|v(\mathcal{E}')\| \leq \mathcal{B}(\mathcal{E}')$, and its ciphertext size and modulus match $K_{\text{size}}(\mathcal{E}')$ and $q(\mathcal{E}')$.

Inductive Step: Let $\mathcal{E}$ be an expression of size $N > 1$. By definition, $\mathcal{E}$ is formed by applying an operation to sub-expression(s) of size strictly less than N. We examine each case:

1. **Negation**: $\mathcal{E} = \mathsf{negate}(\mathcal{E}_1)$. The sub-expression $\mathcal{E}_1$ has size $N - 1 < N$. By the inductive hypothesis, $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$. By Lemma 15 of the SEAL manual, $v_{\text{neg}} = -v(\mathcal{E}_1)$, which implies

$$\|v(\mathcal{E})\| = \|-v(\mathcal{E}_1)\| = \|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1) = \mathcal{B}(\mathcal{E}).$$

2. **Homomorphic Addition and Subtraction**: For $\mathcal{E} \in \{\mathsf{add}(\mathcal{E}_1, \mathcal{E}_2), \mathsf{sub}(\mathcal{E}_1, \mathcal{E}_2)\}$, the sub-expressions $\mathcal{E}_1$ and $\mathcal{E}_2$ have sizes $N_1, N_2 \geq 1$ satisfying $N_1 + N_2 + 1 = N$. Consequently, $N_1 < N$ and $N_2 < N$. By the inductive hypothesis, $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$ and $\|v(\mathcal{E}_2)\| \leq \mathcal{B}(\mathcal{E}_2)$. By Lemma 10 and Lemma 16 of the SEAL manual:

$$\|v(\mathcal{E})\| \leq \|v(\mathcal{E}_1)\| + \|v(\mathcal{E}_2)\| \leq \mathcal{B}(\mathcal{E}_1) + \mathcal{B}(\mathcal{E}_2) = \mathcal{B}(\mathcal{E}).$$

3. **Plain Addition and Subtraction**: $\mathcal{E} \in \{\mathsf{add_plain}(\mathcal{E}_1, m_2), \mathsf{sub_plain}(\mathcal{E}_1, m_2)\}$. The sub-expression $\mathcal{E}_1$ has size $N - 1 < N$. By the inductive hypothesis, $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$. For plain addition, by Lemma 14 of the SEAL manual, the resulting ciphertext $\mathrm{ct}_{\text{padd}}$ has invariant noise $v_{\text{padd}} = v(\mathcal{E}_1) - \frac{r_t(q)}{q}m_2$. Applying the triangle inequality:

$$\|v(\mathcal{E})\| = \|v_{\text{padd}}\| \leq \|v(\mathcal{E}_1)\| + \frac{r_t(q)}{q}\|m_2\| \leq \mathcal{B}(\mathcal{E}_1) + \frac{r_t(q)}{q}\|m_2\| = \mathcal{B}(\mathcal{E}).$$

   Similarly, for plain subtraction, by Lemma 17 of the SEAL manual:

$$\|v(\mathcal{E})\| = \|v_{\text{psub}}\| \leq \|v(\mathcal{E}_1)\| + \frac{r_t(q)}{q}\|m_2\| \leq \mathcal{B}(\mathcal{E}_1) + \frac{r_t(q)}{q}\|m_2\| = \mathcal{B}(\mathcal{E}).$$

4. **Plain Multiplication**: $\mathcal{E} = \mathsf{multiply_plain}(\mathcal{E}_1, m_2)$. The sub-expression $\mathcal{E}_1$ has size $N-1 < N$. By Lemma 13 of the SEAL manual, $\|v_{\text{pmult}}\| \leq \|m_2\|_1 \|v(\mathcal{E}_1)\| \leq N_{m_2}\|m_2\|\|v(\mathcal{E}_1)\|$. Applying the inductive hypothesis $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$:

$$\|v(\mathcal{E})\| \leq \min(\|m_2\|_1,\ N_{m_2}\|m_2\|)\mathcal{B}(\mathcal{E}_1) = \mathcal{B}(\mathcal{E}).$$

5. **Homomorphic Multiplication**: $\mathcal{E} = \mathsf{multiply}(\mathcal{E}_1, \mathcal{E}_2)$. The sub-expressions $\mathcal{E}_1$ and $\mathcal{E}_2$ have sizes $N_1, N_2 < N$ with $N_1 + N_2 + 1 = N$. By the inductive hypothesis, $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$ and $\|v(\mathcal{E}_2)\| \leq \mathcal{B}(\mathcal{E}_2)$. By Lemma 11 of the SEAL manual, the invariant noise satisfies:

$$\begin{aligned}\|v(\mathcal{E})\| \leq & \left[(N_{m_1} + n)\,\|m_1\| + \frac{nt}{2}\frac{n^{j_1+1}-1}{n-1}\right]\|v(\mathcal{E}_2)\| \\ & + \left[(N_{m_2} + n)\,\|m_2\| + \frac{nt}{2}\frac{n^{j_2+1}-1}{n-1}\right]\|v(\mathcal{E}_1)\| \\ & + 3n\,\|v(\mathcal{E}_1)\|\,\|v(\mathcal{E}_2)\| + \frac{t}{2q}\frac{n^{j_1+j_2+1}-1}{n-1}.\end{aligned}$$

Observe that every coefficient on the right-hand side is non-negative: $C_1 = (N_{m_1} + n)\,\|m_1\| + \frac{nt}{2}\frac{n^{j_1+1}-1}{n-1} \geq 0$, $C_2 = (N_{m_2}+n)\,\|m_2\| + \frac{nt}{2}\frac{n^{j_2+1}-1}{n-1} \geq 0$, and $3n > 0$. When the plaintexts m_1, m_2 are unknown, replacing them by their universal upper bounds $\|m_i\| \leq \lfloor t/2 \rfloor$ and $N_{m_i} \leq n$ preserves the validity of the upper bound. Because the bivariate function $f(x, y) = C_1 y + C_2 x + 3nxy + C_0$ is monotonically non-decreasing in both $x \geq 0$ and $y \geq 0$ for non-negative constants $C_1, C_2, C_0 \geq 0$, substituting $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$ and $\|v(\mathcal{E}_2)\| \leq \mathcal{B}(\mathcal{E}_2)$ preserves the inequality:

$$\|v(\mathcal{E})\| \leq \mathcal{B}(\mathcal{E}).$$

6. **Relinearization**: $\mathcal{E} = \mathsf{relinearize}(\mathcal{E}_1)$. The sub-expression $\mathcal{E}_1$ has size $N - 1 < N$ with size $K_{\text{size}}(\mathcal{E}_1) \geq 3$. By the inductive hypothesis, $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$. By Lemma 19:

$$\|v(\mathcal{E})\| \leq \|v(\mathcal{E}_1)\| + \frac{t}{q}(K_{\text{size}}(\mathcal{E}_1) - 2)nBq_\Sigma \leq \mathcal{B}(\mathcal{E}_1) + \frac{t}{q}(K_{\text{size}}(\mathcal{E}_1) - 2)nBq_\Sigma = \mathcal{B}(\mathcal{E}).$$

7. **Galois Rotation**: $\mathcal{E} = \mathsf{rotate}(\mathcal{E}_1)$. The sub-expression $\mathcal{E}_1$ has size $N - 1 < N$ and ciphertext size 2. By the inductive hypothesis, $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$. By Lemma 20:

$$\|v(\mathcal{E})\| \leq \|v(\mathcal{E}_1)\| + \frac{t}{q}nBq_\Sigma \leq \mathcal{B}(\mathcal{E}_1) + \frac{t}{q}nBq_\Sigma = \mathcal{B}(\mathcal{E}).$$

8. **Modulus Switching**: $\mathcal{E} = \mathsf{mod_switch}(\mathcal{E}_1)$. The sub-expression $\mathcal{E}_1$ has size $N - 1 < N$ and size $K_{\text{size}}(\mathcal{E}_1) = j + 1$. By the inductive hypothesis, $\|v(\mathcal{E}_1)\| \leq \mathcal{B}(\mathcal{E}_1)$. By Lemma 21, dropping prime q_k leaves modulus $q' = q/q_k$, with noise:

$$\|v(\mathcal{E})\| \leq \|v(\mathcal{E}_1)\| + \frac{t}{2q'}\frac{n^{j+1}-1}{n-1} \leq \mathcal{B}(\mathcal{E}_1) + \frac{t}{2q'}\frac{n^{j+1}-1}{n-1} = \mathcal{B}(\mathcal{E}).$$

This completes the induction for all operations, establishing that $\|v(\mathcal{E})\| \leq \mathcal{B}(\mathcal{E})$ holds for every well-formed expression of size N.

Proof of Corollaries:

(i) If $\mathcal{B}(\mathcal{E}) < 1/2$, then $\|v(\mathcal{E})\| \leq \mathcal{B}(\mathcal{E}) < 1/2$. By Lemma 1 of the SEAL manual, whenever the invariant noise satisfies $\|v\| < 1/2$, the decryption function correctly removes the noise and outputs the exact message $m(\mathcal{E}) \in R_t$.

(ii) By Definition 2 of the SEAL manual, the invariant noise budget is defined as $\text{Budget} = -\log_2(2\,\|v(\mathcal{E})\|)$. Since $\|v(\mathcal{E})\| \leq \mathcal{B}(\mathcal{E})$ and $-\log_2$ is a monotonically decreasing function:

$$\text{Budget}(\text{ct}_\mathcal{E}) \geq -\log_2(2\mathcal{B}(\mathcal{E})).$$

□

Remark 5 (Straight-Line Programs and Sequences of Length N). An FHE computation expressed as a sequential straight-line program of N instructions:

$$c_1 \leftarrow \mathrm{op}_1(\dots), \quad c_2 \leftarrow \mathrm{op}_2(\dots), \quad \dots, \quad c_N \leftarrow \mathrm{op}_N(\dots),$$

is a special case of the general DAG/AST expression structure. Applying Lemma 22 sequentially to the instruction sequence ensures that each intermediate ciphertext c_k for $1 \le k \le N$ has guaranteed noise bound $\|v(c_k)\| \le \mathcal{B}(c_k)$, and the final result c_N satisfies $\|v(c_N)\| \le \mathcal{B}(c_N)$.

Remark 6 (Implementation in C++). In the reference implementation (`bfv_noise_bound.cpp`), noise estimates are maintained in base-2 logarithmic form $\lambda = \log_2(\mathcal{B})$. Addition and subtraction operations evaluate $\log_2(2^{\lambda_1}+2^{\lambda_2})$ using the helper `log_add`. To prevent numerical under-estimation due to floating-point rounding, all scalar operations apply `round_up` (via `std::nextafter` towards $+\infty$).

For plain addition and subtraction (`add_plain` and `sub_plain`), the C++ implementation returns the input ciphertext bound unchanged ($\mathcal{B}_{\text{out}} = \mathcal{B}_{\text{in}}$). This floating-point simplification is mathematically justified by the parameter regimes: since every valid plaintext satisfies $\|m_2\| \le \lfloor t/2 \rfloor$ and $r_t(q) < t$, the additive remainder term from Lemma 14 is bounded by

$$\frac{r_t(q)}{q}\,\|m_2\| < \frac{t^2}{2q}.$$

In practical BFV parameter sets with $t \in [2^{16}, 2^{20}]$ and $q \ge 2^{100}$, this additive contribution is strictly below 2^{-60} (and often below 2^{-160}). Relative to typical invariant noise levels $\|v\| \in [2^{-30}, 2^{-1}]$, the relative perturbation satisfies $\frac{r_t(q)\|m_2\|}{q\|v\|} \ll 2^{-30}$. This perturbation is well below machine epsilon in IEEE-754 floating-point arithmetic (`long double`), so that numerically

$$\log_2\left(\|v\| + \frac{r_t(q)}{q}\,\|m_2\|\right) = \log_2(\|v\|)$$

holds identically. Consequently, `bfv_noise_bound.cpp` maintains zero noise growth in software without loss of practical fidelity, whereas the mathematical bounds and soundness proofs in this document strictly preserve the unapproximated formula from SEAL manual Lemma 14.